Machine Learning

Exam 1 Review

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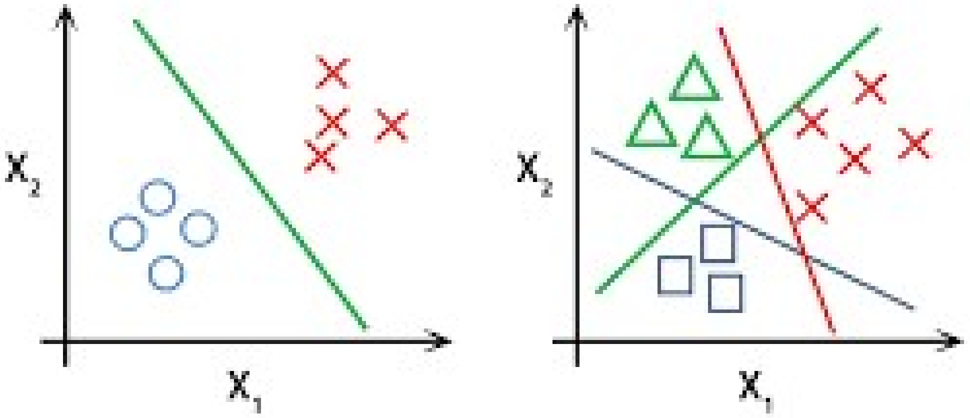
Classification

- Type of problem (binary, multi-class, multi-label)
- Cost functions, loss functions
- Metrics
- Regularization
- NN Output Activations
- Gradient Descent
  - Learning rate
  - Convergence vs Divergence



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A telecom company uses historical usage data to predict whether a specific customer will stay or cancel their subscription next month. What type of problem is this?

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Which of the following best describes a loss function in machine learning?

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A Junior Analyst, "Sam," is tasked with building a model to predict whether a specific customer will stay or cancel their subscription next month. Sam uses a Large Language Model (AI) to generate the code and presents the following model to you for approval before deploying it to production. Do you consider this to be a:

In the provided code, the use of .toarray() to convert a sparse matrix into a dense matrix will reduce the total amount of memory (RAM) consumed by the computer because it simplifies the data structure into a standard array format.

What is wrong with the configuration used by Sam?

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Which metric can you report for this problem?

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Sam is training a neural network in TensorFlow for binary classification. He runs the code here. Is there anything wrong with the setup in the compile method?

Sam got this error. Based on your knowledge of Matrix Algebra and the error message above, why is the code failing?

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A churn prediction model outputs a probability of 0.85 that a customer will churn. The actual label is 1 (the customer churned). What is the binary cross-entropy loss for this single prediction?

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You are given a model that predicts the probability of customer churn for four customers. The actual churn labels (1 = churn, 0 = no churn) and the model's predicted probabilities are shown in the table below. What is the binary cross-entropy cost for this batch?

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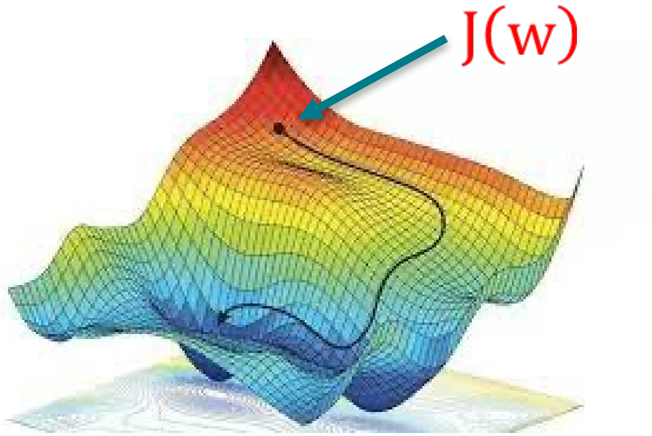
Gradient Descent Algorithm

- Based on going in the direction of the decreasing gradient

$$\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha \nabla J(\mathbf{w})$$



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What is the most likely case given the learning curve given below

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What is the most likely case given the learning curve below

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What is the most likely case given the learning curve of the model fit by Sam

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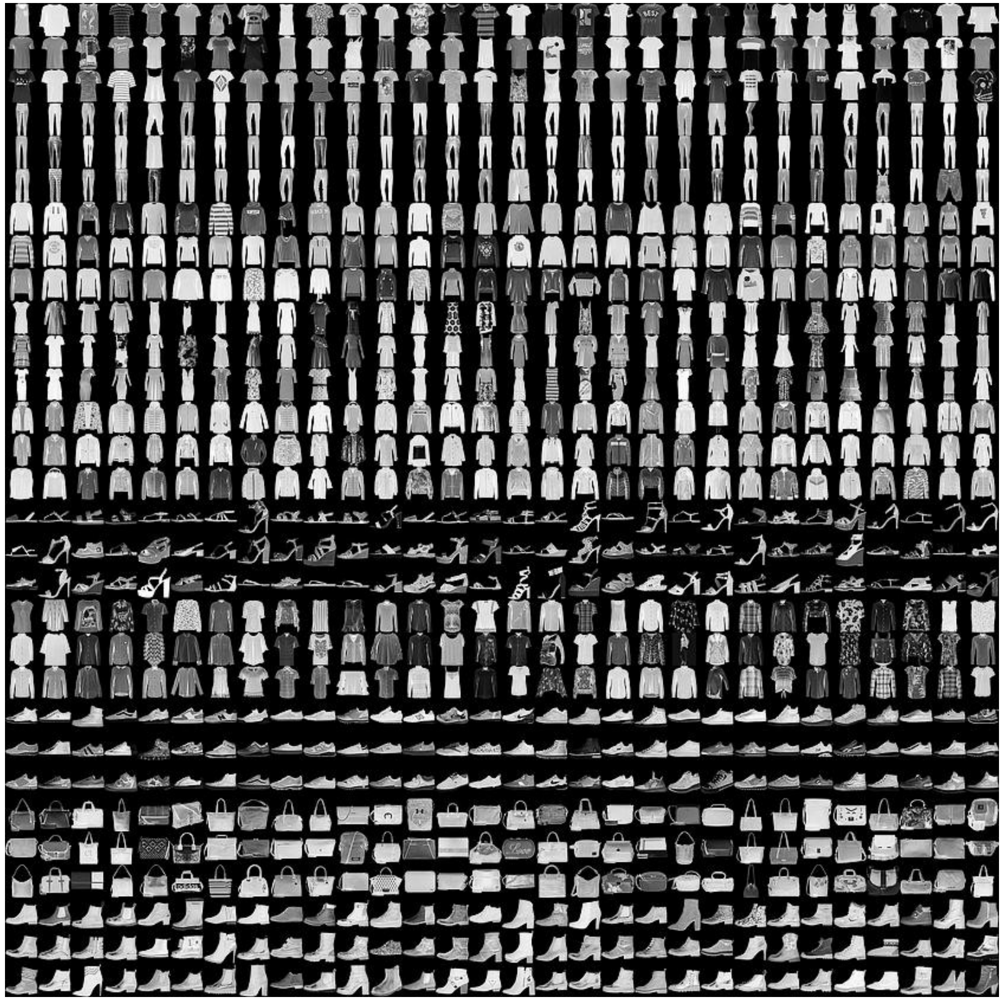
What do you consider the main issue why this model is overfitting?

You are given a model that predicts the probability of customer churn for four customers. The actual churn labels (1 = churn, 0 = no churn). The cost function for binary crossentropy is 0.79.Suppose the model has the following three weights: w = [0.8, -0.3, 1.2], and the regularization parameter alpha = 0.1, what is the total cost: J(w) + alpha*||w||_1

Image Classification Example



Label	Description
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat
5	Sandal
6	Shirt
7	Sneaker
8	Bag
9	Ankle boot

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What type of problem is this?

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What cost function(s) can we use?

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A Fashion MNIST model makes the following probability predictions for an image. The true class is “Sneaker” (Class 7).

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A Fashion MNIST model makes the following probability predictions for 4 images, with the actual class labels provided.

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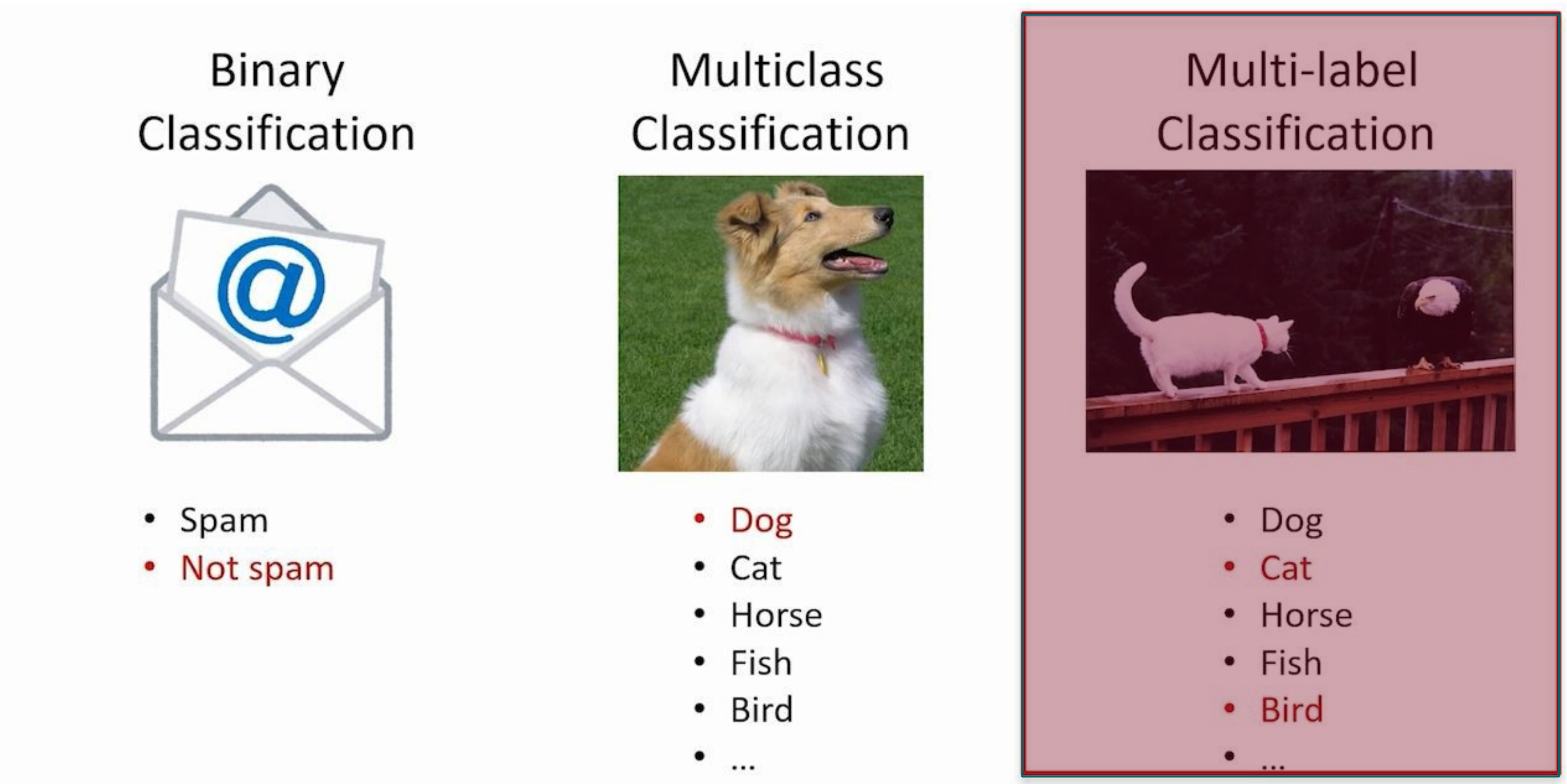
Types of Classification Problems

- The 3 main classification problems are:



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Which statement best describes multilabel classification?

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In a neural network for multilabel classification with 5 possible labels, which output layer is most appropriate?

A multilabel dataset has 200 observations and 7 possible labels. What is the correct shape of the target matrix Y?

Suppose the true labels for one observation are:

[1, 0, 1, 0]

and the predicted probabilities are:

[0.91, 0.35, 0.62, 0.80]

Using a threshold of 0.5, what is the predicted label vector?